

OMANSHU BHATT

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omanshu09.github.io/omanshu-repository

PROFESSIONAL SUMMARY

AI/ML and Full-Stack Engineer building production-grade AI and distributed systems. I architect end-to-end solutions, from retrieval and reasoning pipelines to low-latency APIs and real-time collaborative applications that bridge experimental technology with practical use. My work includes a research-driven web evidence engine, RAG-based code intelligence platform, and CRDT-powered collaborative engineering workspace, where I tackle challenges in information retrieval, realtime synchronization, distributed state, and system reliability. I don't just experiment; I build and ship systems that turn complex technical ideas into reliable, practical software.

TECHNICAL SKILLS

Languages: Python, Go, Java, JavaScript, C

Web & Frameworks: React.js, FastAPI, Node.js, Django, HTML, CSS

Data & Databases: Elasticsearch, PostgreSQL, Redis, Neon

AI / ML: PyTorch, LangChain, Generative AI, RAG, Vector Databases

Cloud, DevOps & Systems: AWS, Docker, Kubernetes, Kafka, CI/CD (Jenkins, GitHub Actions)

Distributed & Realtime Systems: Yjs, CRDTs, Ably, IndexedDB, REST APIs, Distributed State Synchronization

Core CS: Data Structures & Algorithms, Operating Systems, Database Management Systems, Object-Oriented Programming

PROJECTS

Web Research & Evidence Engine <https://search-engine-frontend-xi.vercel.app/>

- Engineered a research-driven web evidence search engine that performs query processing, multi-source web retrieval, content extraction, relevance filtering, ranking to identify high-value sources and aggregates retrieved evidence to detect agreement and contradictions before generating structured, source-backed responses through LLM-based evidence synthesis.
- Architected the end-to-end application with Python/FastAPI services for search orchestration, retrieval, ranking and evidence analysis, a React-based research interface and Dockerized deployment, integrating modular retrieval and processing components into a reliable pipeline for automated web research.

Tech: Web Search, Information Retrieval, Evidence Extraction, RAG, LLM, FastAPI, React, Docker

Benchmarked across 100+ research queries, achieving ~85% relevant source retrieval with sub-10s median response time.

Real-Time Collaborative Engineering Workspace <https://nexus-deployment-frontend.vercel.app/>

- Engineered a real-time collaborative engineering workspace enabling multiple users to edit a shared CRDT-based Notepad with near-instant synchronization, while maintaining an independent Python execution environment and shared execution output; designed the system to remain resilient during connectivity loss through local persistence and automatic state recovery.
- Deployed full-stack system with a React/Vite frontend on Vercel, FastAPI/Node.js backend services on Render, Ably for low latency realtime messaging and synchronization, Yjs CRDTs with IndexedDB persistence for conflict free collaboration and offline durability, Neon PostgreSQL for persistent room state, and E2B isolated sandboxes for secure Python execution.

Tech: React, TypeScript, Vite, Node.js, Express, Yjs, CRDT, Ably, IndexedDB, PostgreSQL, Neon, E2B, REST APIs, Vercel, Render

Instant collaborative synchronization during live multi user editing, with offline local state retention and isolated Python execution through testing

EXPERIENCE

Trainee Intern | Indian Oil Corporation Limited (IOCL)

- Collaborated with cross-functional teams on technical documentation and process optimization.
- Supported data analysis initiatives for operational efficiency improvements, and gained exposure to enterprise-level engineering workflows and quality-assurance practices.

Virtual Intern — AI/ML & Robotics | Kodacy

- Completed two virtual internship tracks covering AI/ML fundamentals, data analysis, and robotics/automation concepts.
- Developed robotics simulation prototypes using Python and relevant frameworks.

CERTIFICATIONS

- Oracle — Agentic AI Certified Foundations Associate
- TCS iON Career Edge - Young professional
- NASA — Hyperspectral Data for Land and Coastal Systems
- Deloitte — Cyber Job Simulation
- Commonwealth Bank — Software Engineering Job Simulation
- University of Oxford — AI, Justice and rule of law

EDUCATION

Bachelor of Technology (B.Tech), Computer Engineering

Govind Ballabh Pant University of Agriculture and Technology, Pantnagar | 2022 – 2026 | CGPA: 7.2

Beersheba Senior Secondary School, Almora

Class XII — 90.4%

Class X — 95.3%